

Phase1 ConfusionBuffer Anki

Phase 1 — Confusion Buffer & Anki Pack (Mathematics + NNs from First Principles)

Companion to the Phase-1 daily schedule (Weeks 3-12). The bedrock; the leaving bar here is the gate that switches the Threads on. (Expanded edition.)

How to use: Block-A derivation → cards + glossary that night → re-derive from a blank page next morning. Prefer explain/derive cards over recall.

Confusion Buffer (tuned for the math build)

Intuition (3B1B) → rigour (MIT/Strang, Stat 110) → derive by hand; glossary nightly; spaced Anki 1/3/7/30 d; Feynman gate = done; stuck >30 min → switch teacher. **Triangulation targets:** SVD-from-eigendecomposition, backprop, matrix calculus, Lagrange multipliers, the reparameterization-free MLE/Bayes glue, attention + $\sqrt{d_k}$.

Ranked hard-topics map

1. **SVD & the spectral picture** — derive from eigendecomposition; PSD; low-rank.
2. **Backprop & autodiff** — the δ -recursion, hand-derived.
3. **Matrix calculus** — gradients of linear layers + softmax-CE.
4. **Probability glue** — Bayes, conditional expectation, $\text{MLE} \leftrightarrow \text{NLL} \leftrightarrow \text{CE}$.
5. **Attention** — scaled-dot-product + $\sqrt{d_k}$ + multi-head.

Anki deck (Q → A)

Deck A · Linear algebra

- **Q:** State the SVD of a real $m \times n$ A. → **A:** $A = U\Sigma V^T$; U, V orthogonal; Σ diagonal $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.
- **Q:** SVD $\leftrightarrow$ eigendecomposition? → **A:** V = eigenvectors of $A^T A$; U = eigenvectors of AA^T ; $\sigma_i = \sqrt{\text{eigenvalues of } A^T A}$.
- **Q:** Why is $A^T A$ PSD? → **A:** $x^T A^T A x = \|Ax\|^2 \geq 0 \ \forall x \rightarrow$ eigenvalues ≥ 0 .
- **Q:** Eckart-Young (best rank-r)? → **A:** $A_r = \sum_{i \leq r} \sigma_i u_i v_i^T$ minimizes $\|A - A_r\|$ in Frobenius & spectral norm.
- **Q:** Geometric meaning of the determinant? → **A:** Signed volume-scaling of the map; $\det=0 \Rightarrow$ collapses dimension $\Rightarrow$ non-invertible.
- **Q:** Rank, column space, null space? → **A:** Rank = # independent columns = $\dim(\text{column space})$; null space = $\{x : Ax=0\}$; rank + nullity = #columns (rank-nullity).
- **Q:** Orthogonal / orthonormal — why nice? → **A:** Orthonormal columns give $Q^T Q = I \rightarrow Q^T = Q^{-1}$, norm-preserving, numerically stable; the basis for QR and stable projections.
- **Q:** Spectral theorem (symmetric matrices)? → **A:** A-real symmetric matrix has real eigenvalues and an orthonormal eigenbasis: $A = Q\Lambda Q^T$.
- **Q:** Least-squares projection of b onto $\text{col}(A)$? → **A:** $x = (A^T A)^{-1} A^T b$; $p = Ax$; residual $b - p \perp \text{col}(A)$ (the normal equations).
- **Q:** Positive-definite vs semidefinite + why care? → **A:** PD: $x^T M x > 0 \ \forall x \neq 0$ (eigenvalues > 0); PSD: ≥ 0 . A PD Hessian $\Rightarrow$ local strict min / local convexity; covariance matrices are PSD.
- **Q:** Condition number — what it tells you? → **A:** $\sigma_{\max}/\sigma_{\min}$; large $\Rightarrow$ ill-conditioned (sensitive to noise, slow/zig-zag gradient descent).
- **Q:** Quadratic form + its gradient? → **A:** $f(x) = x^T A x$; $\nabla f = (A + A^T)x = 2Ax$ for symmetric A.
- **Q:** Pseudo-inverse — what for? → **A:** $A^+ = V\Sigma^+ U^T$ gives the min-norm least-squares solution when A isn't invertible/square.

Deck B · Calculus & matrix calculus

- **Q:** Gradient vs directional derivative? → **A:** The gradient points in the steepest-ascent direction; the directional derivative along u is $\nabla f \cdot u$.
- **Q:** $\partial L / \partial w$ for $z = wx + b$ (shape)? → **A:** δx^T where $\delta = \partial L / \partial z$; same shape as W (outer product). $\partial L / \partial b = \delta$.
- **Q:** $\partial(\text{softmax} + \text{CE}) / \partial \logits$? → **A:** $p - y$ (predicted probs minus one-hot) — the clean result behind softmax-CE.
- **Q:** Jacobian vs Hessian? → **A:** Jacobian = matrix of first partials (vector $\rightarrow$ vector); Hessian = matrix of second partials (curvature of a scalar function).
- **Q:** Chain rule for $f(g(x))$? → **A:** $df/dx = f'(g(x)) \cdot g'(x)$; for vectors, multiply Jacobians right-to-left.
- **Q:** Taylor expansion (to 2nd order) — why useful? → **A:** $f(x + \Delta) \approx f(x) + \nabla f^T \Delta + \frac{1}{2} \Delta^T H \Delta$; the basis for Newton's method and analyzing curvature/optimization.
- **Q:** Convexity — the Hessian test? → **A:** f is convex iff its Hessian is PSD everywhere $\rightarrow$ any local min is global (why convex problems are "easy").
- **Q:** Lagrange multipliers — the idea? → **A:** To optimize f subject to $g=0$, set $\nabla f = \lambda \nabla g$ (gradients parallel at the optimum); λ = sensitivity of the optimum to the constraint.

Deck C · Probability

- **Q:** Bayes' theorem + terms? → **A:** $P(\theta|D) = P(D|\theta)P(\theta)/P(D)$: posterior $\propto$ likelihood $\times$ prior / evidence.
- **Q:** Law of total probability / expectation? → **A:** $P(A) = \sum P(A|B_i)P(B_i)$; $E[X] = E[E[X|Y]]$ (tower rule).
- **Q:** $E[X|Y]$ as a projection? → **A:** The best mean-square predictor of X from Y = orthogonal projection of X onto functions of Y.
- **Q:** E and Var of a sum of independent vars? → **A:** $E[\sum X_i] = \sum E$ (always); $\text{Var}[\sum X_i] = \sum \text{Var}$ (only if independent/uncorrelated).
- **Q:** Independent vs uncorrelated? → **A:** Independent $\Rightarrow$ uncorrelated; the converse fails in general (uncorrelated just means zero linear association). For jointly Gaussian, they're equivalent.
- **Q:** Name 4 distributions + a use? → **A:** Bernoulli/Binomial (counts of successes), Poisson (rare-event counts), Gaussian (CLT/noise), Exponential (waiting times).
- **Q:** Key Gaussian facts? → **A:** Fully specified by mean+covariance; linear combos of Gaussians are Gaussian; marginals/conditionals are Gaussian; max-entropy for fixed variance.
- **Q:** What is a covariance matrix + a property? → **A:** $\Sigma_{ij} = \text{Cov}(X_i, X_j)$; symmetric PSD; its eigenvectors are the PCA axes.

- **Q:** Why is minimizing NLL = MLE = minimizing CE? → **A:** $NLL = -\sum \log p(x|\theta) \rightarrow$ maximizing log-likelihood (MLE); for a categorical target it is the cross-entropy between the label and the prediction.
- **Q:** Conjugate prior — why care? → **A:** A prior whose posterior stays in the same family (e.g., Beta-Bernoulli) → closed-form Bayesian updates.

Deck D · Neural networks from scratch

- **Q:** Backprop in one sentence? → **A:** The chain rule applied layer-by-layer, reusing cached forward activations to get $\partial L / \partial \text{param}$ for all params in one backward pass.
- **Q:** Backward one layer: δ_{l-1} ? from δ_l ? → **A:** $\delta_{l-1} = (W_l^T \delta_l) \odot f'(z_{l-1})$.
- **Q:** What is a computational graph? → **A:** A DAG of elementary ops; forward computes values, backward applies the chain rule in reverse-topological order accumulating gradients.
- **Q:** Reverse-mode autodiff — cost? → **A:** Exact gradients at $\approx$ one forward pass of compute (memory for cached activations) → why it's used for scalar losses of many params.
- **Q:** Vanishing gradients — cause + 2 fixes? → **A:** Repeated $\times$ small activation-derivatives shrinks gradients with depth; fixes: ReLU-family + residual connections + normalization + good init.
- **Q:** Exploding gradients — fix? → **A:** Gradient clipping (by norm) + proper init + normalization.
- **Q:** Xavier vs He init — why? → **A:** Scale init so activation/gradient variance is preserved across layers — Xavier (tanh/linear), He (ReLU); prevents exploding/vanishing at the start.
- **Q:** Activation comparison (sigmoid/tanh/ReLU/GELU)? → **A:** sigmoid/tanh saturate (vanishing gradients); ReLU is fast, non-saturating but can die; GELU/Swish are smooth ReLU variants used in transformers.
- **Q:** Why residual connections? → **A:** $y=x+F(x)$ gives gradients a direct path (identity) → trains very deep nets and eases optimization (mitigates degradation).
- **Q:** What does BatchNorm do / why help / when break? → **A:** Normalize a feature over the batch then scale/shift (γ, β); smooths the landscape, enables higher LR; breaks with tiny batches, needs running stats at eval.
- **Q:** Softmax + cross-entropy — why combined numerically? → **A:** Combine into log-softmax/logsumexp for stability (avoid overflow/log(0)); gradient wrt logits = $p - y$.

Deck E · Attention (end of phase)

- **Q:** Scaled-dot-product attention? → **A:** $\text{softmax}(QK^T/\sqrt{d_k})V$.
- **Q:** Why $\div \sqrt{d_k}$? → **A:** QK^T has variance $\sim d_k$; rescaling to unit variance stops softmax saturating into near-one-hot (which vanishes gradients).
- **Q:** Q, K, V intuition? → **A:** Query = what I seek; Key = what I offer; Value = what I pass if matched; output = similarity-weighted sum of values.
- **Q:** Why multi-head? → **A:** Parallel lower-dim heads capture different subspaces/relations, concatenated — more expressive than one head.
- **Q:** Self- vs cross-attention? → **A:** Self: Q,K,V from the same sequence (intra-sequence relations); cross: Q from one sequence, K,V from another (e.g., decoder attending to encoder).

Common misconceptions & traps

- **Backprop isn't magic** — it's the chain rule + caching; hand-derive it for a 2-layer MLP.
- **BatchNorm behaves differently at train vs eval** (batch stats vs running stats) — a classic bug.
- **Initialization matters** — poor init causes exploding/vanishing signals before training starts.
- **Depth without residuals/normalization doesn't help** — signal degrades; that's what residual/norm fix.
- **SVD is not just PCA** — PCA is SVD on the *centered* data; SVD is the general factorization.
- **Uncorrelated $\neq$ independent** (except jointly Gaussian).
- **A large condition number** silently slows/destabilizes gradient descent — normalize/precondition.

Glossary starter

SVD / singular value · eigenvector/eigenvalue · rank/null space/rank-nullity · orthonormal/QR · spectral theorem · PSD/PD · Eckart-Young · determinant · projection/least-squares/normal equations · condition number · quadratic form · pseudo-inverse · gradient/directional derivative · Jacobian/Hessian · chain rule · Taylor expansion · convexity (PSD Hessian) · Lagrange multipliers · Bayes · law of total probability/expectation (tower) · conditional expectation · independent vs uncorrelated · Bernoulli/Binomial/Poisson/Gaussian/Exponential · covariance matrix · MLE/MAP/NLL/CE · conjugate prior · backprop (δ -recursion) · computational graph · autodiff (reverse mode) · vanishing/exploding gradients · gradient clipping · Xavier/He init · ReLU/GELU/dead-ReLU · residual connection · BatchNorm (train/eval) · log-softmax/logsumexp · scaled-dot-product attention · $\sqrt{d_k}$ · Q/K/V · multi-head · self/cross-attention.

Drills

Whiteboard: derive SVD from the eigendecomposition of $A^T A$; prove $A^T A$ is PSD; least-squares normal equations; $\partial(\text{softmax} + \text{CE}) / \partial \text{logits} = p - y$; Lagrange-multiplier setup for a constrained optimum; full backprop for a 2-layer MLP; scaled-dot-product attention + $\sqrt{d_k}$. **Blank-file:** SVD-based PCA + truncated-SVD compression; a subspace projection; softmax+CE fwd/bwd with a numerical gradient check; **micrograd** (scalar autodiff + MLP) from memory; multi-head attention; a residual block; He-init a small net and check activation variances.

Leaving bar (cold, no notes) → Threads T/M/R begin

Derive SVD from eigendecomposition + why $A^T A$ is PSD; the least-squares normal equations; $\partial(\text{softmax} + \text{CE}) / \partial \text{logits}$; hand-derive backprop for a 2-layer MLP + implement micrograd from memory; explain vanishing/exploding gradients + fixes; why BatchNorm helps + what breaks; Bayes + $E[X|Y]$ -as-projection + $\text{MLE} \leftrightarrow \text{NLL} \leftrightarrow \text{CE}$; whiteboard scaled-dot-product attention with $\sqrt{d_k}$ + multi-head.